

ML WITH PYTHON | FINAL PROJECT

Predicting New-User Engagement on Zindi

Will a user who just signed up still be active in their second month?

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ML with Python, Final Project

Data: Zindi New User Engagement Prediction Challenge

Roadmap

1. Problem: why it matters, how ML helps
2. Data: load with pandas, time axis, target, X / y
3. Data preparation
4. Exploratory analysis (Seaborn)
5. Validation, metric, models, tuning
6. Results, impact, conclusions

Two label definitions

The M4 activity log ends on day 22.
All models are trained and tested with **two labels**:

- **A**: active in the whole next month
 - **B**: active in days 1–22 of the next month (aligned)
-

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SECTION 1

Problem

1. Problem
2. Data and target
3. Data preparation
4. Exploratory analysis
5. Validation, metric and models
6. Results and impact

The problem and why it matters

Context

- Zindi: Africa's largest data-science competition platform, ≈ 12 k sign-ups in 7 months
- Only **12–21%** of a typical cohort comes back in month 2

Task

- At the end of a user's **first month**, predict: active in the **next month**?
- Binary classification, minority positive class

Why it matters

- Acquisition is expensive; month-2 survivors become competitors and hires
- Churn is fast: Zindi must act right after month 1

How the model helps

- **Lower cost:** retention offers only to users who need them
- **Prioritise:** invite likely-engaged users to hackathons
- **Insight:** which behaviours drive retention

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SECTION 2

Data and target

1. Problem
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Loading the data and recovering the time axis

- 8 tables loaded with `pandas.read_csv` (Users, UserActivity with 317 k events, ...)
- Months are **masked**; sign-up $\times$ activity table is upper-triangular only in the order
M11 $\rightarrow$ M12 $\rightarrow$ M1 $\rightarrow \dots \rightarrow$ M5
- M4 cohort = official test set (M5 activity removed)

X: 9 026 users $\times$ 50 features
 y_A, y_B : Active (0/1)

Events by sign-up cohort $\times$ activity month

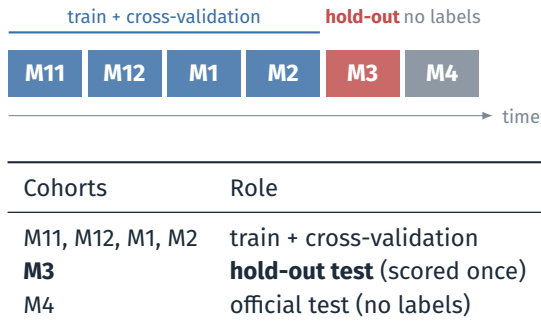
Sign-up cohort	M11	15,157	3,375	2,465	1,235	1,317	503	608
	M12	0	38,653	24,299	7,468	3,012	1,448	2,517
	M1	40	14	31,489	7,041	3,567	2,243	1,740
	M2	0	0	0	45,428	9,676	2,701	3,377
	M3	0	0	0	0	38,662	5,183	3,177
	M4	28	0	32	0	74	14,880	379
	M5	4	5	7	9	15	4	45,460
		M11	M12	M1	M2	M3	M4	M5

Activity month

red box: removed by Zindi = labels of the official test cohort

Test on a period later than the training data

- Time-ordered data $\Rightarrow$ the test set must come **after** the training data
- Labelled cohorts: M11, M12, M1, M2, M3
- $\Rightarrow$ the **latest** one, **M3**, is our hold-out test cohort
- Any other choice trains on the future or wastes a month of data

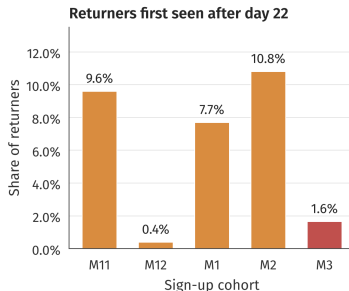
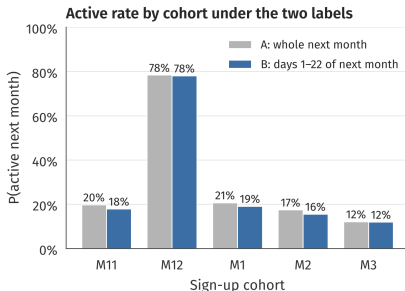


Truncated labels in the hold-out cohort

Table	Last day in M4
UserActivity*	22
CompetitionPart.	30
Discussion	30

*page views and clicks, most of the events

- M3's label is measured in M4
- Users returning after day 22 → labelled **0**
- Normally 8–11 % of returners come after day 22; in M3 only **1.6 %**



Label-definition inconsistency and label B

With a whole-month label (A)

- Training: $y = 1$ = “active in the next ~30 days”
- Test (M3): $y = 1$ = mostly “active in the first 22 days”
- Features aligned, **labels not aligned**

Consequences

- Biased F1 estimate on M3
- Threshold tuned on 17–21% return rates, applied to an artificially low cohort
- Lower M3 return rate is partly an artefact of the cutoff

Label B

Active = any record in **days 1–22** of the next month, for **every** cohort, training and test.

Same features, folds and models.
We run everything with **A** (original) and **B** (aligned) and report both.

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SECTION 3

Data preparation

1. Problem
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Preparing the dataset

Issue	What we do	Why
Multi-table event logs	aggregate per user, sign-up month only	one row per user, no leakage
70+ event titles	18 behaviour families	type of behaviour generalises
Country missing (47 %)	Missing category + flag	missingness is informative
146 countries	top 10 + Other, one-hot	avoids over-fitting rare levels
FeatureY codes, "count 10" strings	one-hot; parse number, NaN $\rightarrow$ 0	nominal codes; NaN = none
Late sign-ups have less time	days_observed, active_day_ratio	normalise by exposure

New features: recency (days_since_last), regularity (n_active_days), behaviour counts, joined_hackathon, n_new_comps_joined, discussions, profile.

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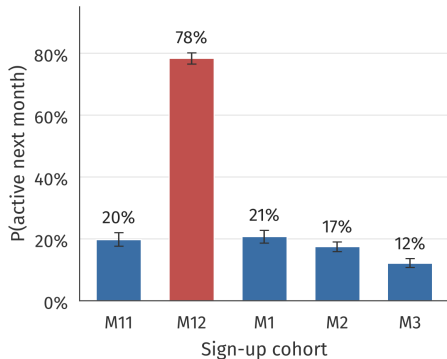
SECTION 4

Exploratory analysis

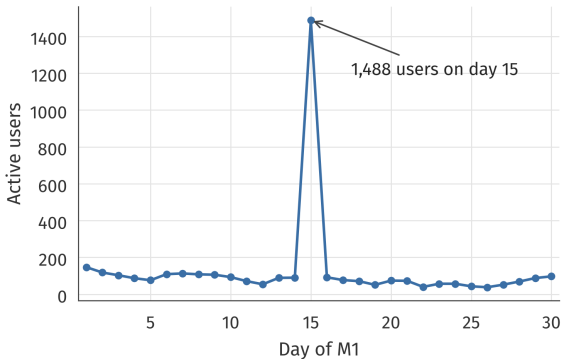
1. Problem
2. Data and target
3. Data preparation
- 4. Exploratory analysis**
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EDA 1: retention by cohort

Share active in month 2, by cohort

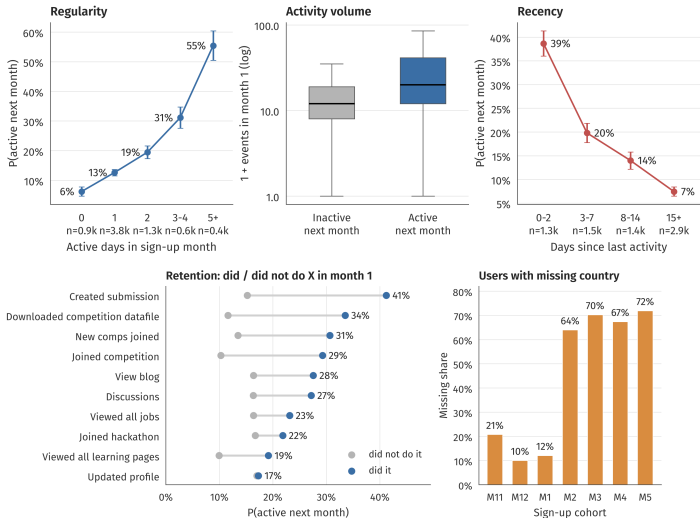


M12 cohort: daily active users in M1



- M12 retains **78 %**: one hackathon spike (≈ 1500 users on one day)
- $\Rightarrow$ **time-based validation**; test **dropping M12** from training; add `joined_hackathon`

EDA 2: first-month engagement and behaviour



- Active days: 6 % → 55 % return
- Seen in last 2 days: 39 % vs 7 %
- Submitting / downloading ≫ reading blogs
- Country missing jumps in M2 (form change) ⇒ test **dropping country**
- Skewed, collinear counts ⇒ log + L2 logistic, trees

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SECTION 5

Validation, metric and models

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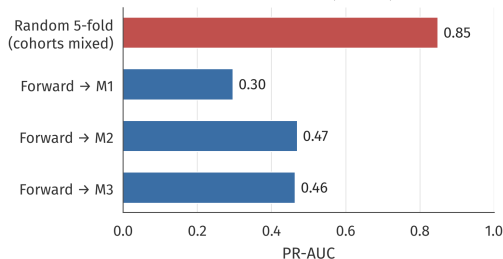
Cross-validation and metric

Forward chaining by cohort

Split	Train	Validate
Fold 1	M11 (+M12)	M1
Fold 2	M11 (+M12), M1	M2
Hold-out	all above	M3

Metric: F1 of the positive class (competition metric; accuracy useless at 12 % positives).
Model selection by **PR-AUC**; threshold tuned on out-of-fold predictions.

Random K-fold vs. forward (label A)



Random K-fold reports $\sim 2\times$ higher PR-AUC because it leaks cohort information.

Models and hyper-parameter search

Models

- **Baseline:** active on $\geq k$ days
- **Logistic regression** (log, L2)
- **Random forest**
- **Gradient boosting**

GridSearchCV on our forward folds, crossed with: keep / drop M12, keep / drop country. Run once per label.

CV PR-AUC, M12 dropped, no country

Model	Label A	Label B
GradBoost	0.508	0.495
RandomForest	0.498	0.488
LogReg	0.474	0.467
GradBoost, M12 kept	0.425	0.392

Dropping M12 helps every model. Both labels select **GradBoost**.

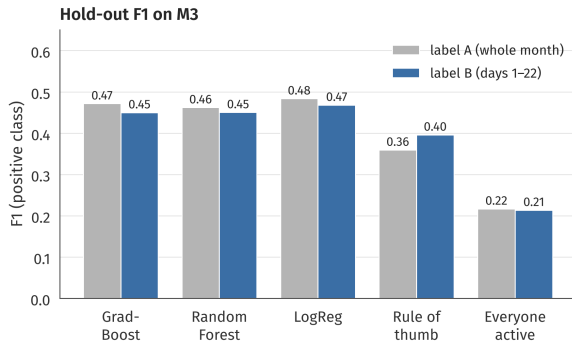
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SECTION 6

Results and impact

1. Problem
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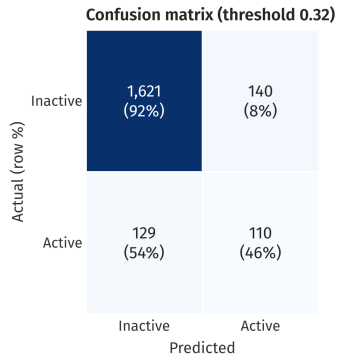
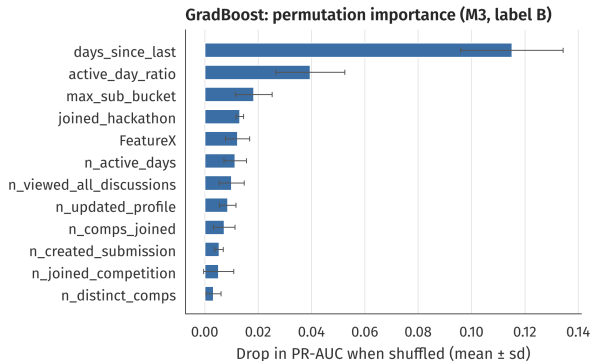
Hold-out results on M3: label A vs. label B



GradBoost on M3	A	B
F1	0.47	0.45
Precision	0.45	0.44
Recall	0.49	0.46
ROC-AUC	0.83	0.83
Best rule of thumb F1	0.36	0.40

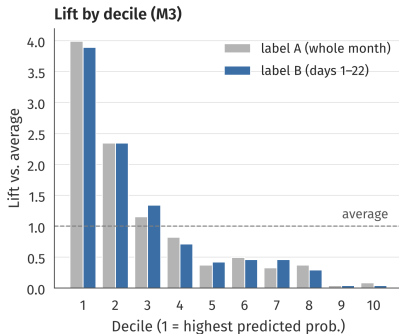
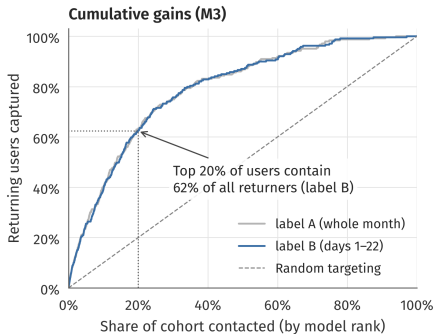
- A mixes two label definitions
- **B is the consistent estimate**
- Gap ≈ 0.02 : within noise, same conclusions

Feature importance



- **Recency** first, then regularity and active behaviour (submissions, hackathons); demographics add little
- Finds 46 % of returners at 44 % precision ($\approx 3.7\times$ the base rate)

Real-world impact



- Top 20 % of users hold **62 %** of returners
- Top decile: $\sim 4\times$ the average
- Causal effect needs an **A/B test**

Example: 2 000 sign-ups, 5 USD each
all users: 10 000 USD $\rightarrow$ middle 40 %:
4 000 USD (**-60 %**)

Conclusions

Findings

- Recovered month order; tested on the latest cohort
 - Fixed a label-definition inconsistency (day-22 cutoff)
 - Recency and regularity are the main predictors; active behaviour matters more than browsing
 - Non-stationary data: validation design mattered more than the model
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Does it solve the problem?

- **Ranking: yes**; exact classification: partly
 - F1 0.45 (consistent) vs 0.40 rule of thumb
 - Targeted retention $\approx 60\%$ cheaper
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Thank you